



M359 Scenarios

These scenarios have been written specifically to support the course *M359 Relational databases: theory and practice*. Consequently, they have a number of simplifications that will probably make them unlike any real world examples you may have encountered. However, they do represent situations in which database development would be required and the content should be sufficiently sensible that any divergence from the real world will not distract from the use of these scenarios.

You should note the following details.

- ▶ The scenarios present the data requirements, not the processing requirements for that data.
- ▶ You should not attempt to bring any 'real-world' knowledge of similar situations that you are familiar with. Consider that the scenario and related documents have been produced by a data analyst after their exploration of a specific situation.
- ▶ If there are aspects of the model you think may be incorrect then discuss these with other students, via the FirstClass conferences, or with your tutor.

The scenarios should be read in conjunction with the relevant domain of discourse summary, which give further background to these scenarios. You are most likely to require this database card when working through Section 5 of *Block 1*.

M359 Hospital scenario

The hospital is organised into a number of wards, each of which may be empty or may be occupied by one or more patients. Each ward is identified by a ward number; it has a name and contains a fixed number of beds. Each ward is staffed by one or more nurses. Each nurse is identified by a staff number and has a name. A nurse is assigned to a single ward. Some nurses are designated to supervise one or more other nurses on the same ward. A nurse has at most one supervisor.

Each patient in the hospital has a patient identification number and name recorded, their gender, height and weight is kept on record. Each patient is assigned to a single ward and is under the care of (is the responsibility of) a single doctor who must have a position of consultant. Each consultant is responsible for a number of patients. Details of the junior doctors in the hospital, who have a position of either registrar or house officer, are also recorded. Consultants must have a specialism, but registrars and house officers are not specialists and do not have a specialism. All specialists are consultants. Each doctor (whether consultant or junior doctor) in the hospital has a staff number and name recorded. Each consultant must head a single group, known as a team, consisting of one or more house officers and registrars. A team must be headed by a consultant. A house officer or registrar cannot head a team, but must be a member of one team, and consultants cannot be members of a team. Each team has a team code and a telephone number that people can use to leave messages for the team.

Each doctor may provide treatment for several patients and each patient may receive treatment from a number of doctors. Any doctor treating a patient must be a member of the same team as the consultant responsible for that patient. A patient may receive more than one treatment from each doctor, for which the start date and the reason (e.g. a chest infection) for the treatment are recorded. Such a treatment may require several prescriptions, each of which has a prescription number and specifies a total quantity and daily dosage of a drug. A drug may appear on different prescriptions. Each drug has a unique code, a type and unit price as well as a name (e.g. Dolensol or Hallexanuran).

M359 University scenario

An open learning university needs to keep details of its staff and students, the courses that are currently available and the performance of the students currently enrolled on those courses. Only information about the current year is to be recorded.

Staff and students are managed by the Regional Offices. The Regional Offices are known by their reference numbers, 1 through 12; each office has an address, a telephone contact number and an email address.

Information about each student is initially recorded at registration. This includes the student's identification number (issued at the time the student registers), name, address and date of registration. Each student is managed by one region, initially the region in which they register but this may change. A student is not required to enrol on any courses at registration; enrolment on a course can happen at some later time but the date of that enrolment is required. An email address is recorded for each student (if they do not have one at registration, then the University gives them a student email account).

Information recorded for each member of staff includes the staff number, name, address, telephone number and one email address. In addition, the name of the region that employs the staff member is important. Each staff member may or may not act as a tutor to one or more students on courses, and may or may not act as a telephone tutor on a course. A staff member can only be a telephone tutor for a course if they are tutoring at least one student on the same course.

Each student has a tutor for each course on which they are enrolled. The allocation of a student's tutor for a course takes place sometime after enrolment, and there may be times when a student does not have a tutor for a course on which they have enrolled. A staff member may only tutor students who are managed by the same Regional Office.

Each course which is available for study is given a course code, a title and a value for credit points – either a 30-point course or a 60-point course. A course may have a quota – the maximum number of students that can be enrolled on the course in any one year; each quota is reviewed regularly and the last date of review is recorded with the quota. A course need not (yet) have any students enrolled on it. Students may not enrol for more than 180 points' worth of courses at any one time. A course must have at least one staff member who acts as a telephone tutor for that course.

Assessment on the course is by assignment and examination. A course has, at most, one examination which always takes place at the end of the course. A record of the examination centre used by each student is required. A 60-point course may have up to five assignments associated with it, and a 30-point course up to three assignments. The mark awarded to a student on an individual assignment or for an examination on a given course is recorded as a percentage. A student can submit, along with an assignment or following the examination, a special circumstances form whose form reference number and the date on which the form is received will be noted by the examination officer. Only one form per assignment and one form per examination are permitted; each form can include details of only one assignment or one examination.
